

Elizabeth Porter, MSc, PhD Candidate

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<https://elizabeth-porter.github.io/>

LIFETIME CURRICULUM VITAE

EDUCATION

PhD Candidate, 2022 – present. **Status:** In progress. Anticipated completion: April 2027.

University of Guelph, Department of Integrative Biology.

Advisors: Dr. Karl Cottenie (Principal advisor), Dr. Shoshanah Jacobs (Co-advisor),
Dr. Adam Smith (Committee member), Dr. Shelby Bohn (Committee member)

Thesis: *Advancing the Appropriate Use of Citizen Science Data*

MSc, 2020 – 2022 [GPA: 4.0/4.0]

University of Guelph, Department of Integrative Biology.

Advisors: Dr. Karl Cottenie (Principal advisor), Dr. Shoshanah Jacobs (Co- advisor),
Dr. Lisa Robertson (Committee member)

Thesis: *Squirrel Life: Citizen Science as an Accessible Framework for Experiential Learning in Biology*

BSc, Honours with Distinction, Bio-Medical Science and Zoology, 2014 – 2018 [GPA 3.9/4.0]

University of Guelph, Department of Biomedical Sciences

Major Research Project: *Aerobic Exercise as a Treatment for Major Depressive Disorder*

Advisor: Dr. Genevieve Newton

Major Research Project: *Does rearing environment influence the effect of single nucleotide polymorphisms on the genetics of skeletal traits in laying hens?*

Advisor: Dr. Gregoy Bédécarrats

RESEARCH EXPERIENCE

2023-2025 Manufactured Ecosystems **Project Manager**

- Coordinate a team of 50+ international collaborators including researchers, artists, & writers.
- Contributed to research conceptualization, analysis, and dissemination. Led and organized multiple seminars and workshops, engaging >100 international researchers, artists, and community members across disciplines including environmental science, engineering, literature and education. Co-curated an art exhibit featuring interactive installations and prototypes exploring climate adaptation through nature-inspired technologies.

2023-2025 Climate Stories **Research Associate and Supervisor**

Guelph Institute for Environmental Research

- I began as a research associate, developing study design and conducting interviews. In 2025 I transitioned to research supervisor overseeing analyses and project completion. Working with researchers and creative practitioners from arts and sciences, we examine how storytelling shapes climate understanding, knowledge translation, and public engagement using a bottom-up approach.

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2022

Squirrel Mapper Project Collaborator

Guelph, ON

- Organized site selection, deployment and retrieval of camera traps (15 sites), and weekly point count surveys (15 sites) around Guelph. <https://www.squirrelmapper.org/our-team>

2020-2022

Squirrel Life Founder and Project Manager

- Transdisciplinary research and outreach initiative developed as part of my MSc research to promote equitable access to science education. I led all stages of the project from concept to implementation, including developing a mobile app, and secured \$34,700 in funding. I led a team of students, faculty, and community partners, strengthening my research autonomy, critical judgement, and leadership in collaborative scholarship. Integrated into BIOL*1070 curriculum and used by >9,000 students since 2020. Received MEDIA COVERAGE across 11 news outlets and a podcast (2021).
- Hosted a public nature event at the University of Guelph Arboretum with over 200 attendees.
- Developed novel citizen science data collection protocols to record behavioural observations of various squirrel species in diverse environments including urban green spaces, residential and commercial areas, and rural and remote areas.
- Piloted protocols with over 2000 volunteer participants from multiple cities in North and South America, Europe, Africa, and Asia.
- Managed development of mobile application and produced inclusive and accessible protocols. This component lays the foundations for a global initiative to change the way citizen science data is collected and humanize Big Data, by integrating ecological, geospatial, and sociodemographic data.

2019-2020

Research Technician

Centre for Biodiversity Genomics, University of Guelph

- Insect specimen identification (Family level), and laboratory techniques to prepare specimens for DNA barcoding and long-term storage.
- Organize and participate in Natural History collection research trips to review available specimens and select an assortment for DNA barcoding and cataloguing.
- Assistant to the Herbarium Plant Lead Research Associate.
- Extensive database management and record creation.
- Developed protocols and conducted insect trapping (Malaise, Pitfall, Lindgren funnel, Pan).

2018

Laboratory Technician

Reproductive Health and Biotechnology Lab, Ontario Veterinary College, Guelph ON

- Assessed quality and harvested bovine reproductive tracts and prepared them for use in health and cancer research projects.

2017

Research Assistant

School of Environmental Sciences, University of Guelph

- Developed research and data collection protocols and conducted crop damage assessment at 10+ field sites across Ontario. Collected and maintained insect colonies. Designed experiments and researched sustainable Integrative Pest Management practices.
- Assisted graduate students with designing and running experiments, data interpretation, and creative problem solving to overcome obstacles and improve workflow.

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PUBLICATIONS

Eight peer-reviewed publications; Four forthcoming.

Refereed Publications

Invited Publications

- 8 Jacobs, S., **Porter, E.**, Dowhaniuk, D., Lipton, M., Bazely, D., Smith, M. A., Zampaki, N., Karpouzou, P., Eggermont, M., Helms, M., Summers, M., Jones, A., Smylitopoulos, C., Clitheroe, H., Davies, A., Hinds Myrie, M., Gillis, D., Rivera, C., Benessaiah, K., Wanieck, K. (2024). Can Manufactured Ecosystems fully replace ecosystem services? *Zygote Quarterly*, 35, 40-53. <https://zqjournal.org/editions/zq35.html>

Journal Articles

- 7 Myrie, M., Bohn, S., Waniek, K., Gillis, D., Jones, A., Clitheroe, H., Davies, A., **Porter, E.**, Dowhaniuk, D., Smylitopoulos, C., Benessaiah, K., Smith, M. A., Bazely, D., Jacobs, S. (In Press). *Building and Sustaining Transdisciplinary Research Teams for Environmental Justice: Lessons from the Manufactured Ecosystems Project. Equity in Education & Society.*
- 6 Jacobs, S., Zhang J., Wolf, E., **Porter, E.**, Bohn, S., Sparks, A., Eggermont, M., Summers, M., Rivera, C. I. R., Clitheroe, H., Gillis, D., Smith, M. A., Benessaiah, K., Jones, A., Davies, A., Helms, M., Bazely, D., Lipton, M., Dowhaniuk, D., van Vierssen Trip, N., Zampaki, N., Karpouzou, P., Wanieck, K. (2025). Are ecosystem services replaceable by technology yet? Bio-inspired technologies for ecosystem services: Challenges and opportunities. *Biomimetics*. 10(11), 784. <https://doi.org/10.3390/biomimetics10110784>
- 5 Wanieck, K., Smith, M. A., **Porter, E.**, Zhang, J., Dowhaniuk, D., Jones, A., Gillis, D., Lipton, M., Hinds Myrie, M., Bazely, D., Eggermont, M., Summers, M., Smylitopoulos, C., Rivera Cardenas, C. I., Wolf, E., Karpouzou, P., Zampaki, N., Clitheroe, H., Davies, A., Castillo, A. H., Helms, M., Benessaiah, K., Jacobs, S. (2025). A call for bio-inspired technologies: Promises and challenges for ecosystem service replacement. *Biomimetics*, 10(9), 578. <https://doi.org/10.3390/biomimetics10090578>
- 4 **Porter, E.**, Beltran, M., Jacobs, S. (2022). Curricular Nature-Based Learning in Higher Education to Support Mental and Environmental Health. *Health Science Inquiry*, 13, 98-107. <https://healthscienceinquiry.com/index.php/hsi/issue/view/14/15>
Awarded best submission of the year.
- 3 Robertson, L., **Porter, E.**, Smith, M. A., Jacobs, S. (2021). Evidence-based course modification to support learner-centered and student-driven teaching in a pandemic: Leveraging digital and physical space for accessible, equitable, and motivating experiential learning and scientific inquiry in a first-year biology course. *International Journal of Higher Education*, 10(7), 96-109. <https://doi.org/10.5430/ijhe.v10n7p96>.

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Conference Publications (Peer-reviewed)

- 2 **Porter, E.**, Luo, M. X., Lit, B., McKechnie, I., Saha, J., Ratra, P., Lewis, N., Norman, Z., Cottenie, K., Jacobs, S., Gillis, D. (2022). Developing a transdisciplinary citizen science tool for experiential learning in undergraduate education: Squirrel Life in a nutshell. *16th International Technology, Education and Development Conference*. doi: [10.21125/inted.2022.0535](https://doi.org/10.21125/inted.2022.0535)
- 1 Bianco, A., Morey, K., Arbour, K., Avey, S., Gu, C., Li, J., Lin, J., Lin, K., Oaks, C., **Porter, E.**, Zehr, L., Horner, K., Bock, J. (2019). Unlocking the untapped potential of volunteers: A contemporary re-imagining of the volunteer program. *International Association of Avian Trainers and Educators Conference*. Orlando, Florida.

Forthcoming Journal Articles

- 4 Porter, E., Cottenie, K., Jacobs, S. *Citizen Science is a Scalable Experiential Activity for Whole Person Learning in Post-Secondary Education*. [Under review at *BioScience*].
- 3 Porter, E., Cottenie, K., Smith, A. C., Jacobs, S. *Unsung Scientists*. [Submitted to *Frontiers in Ecology and the Environment*].
- 2 Porter, E., Wise, C., Cottenie, K. Quantifying Observer Variation in a Small-Scale Volunteer Bird Survey. [Awaiting final coauthor feedback].
- 1 Lewis, N., Cottenie, K., **Porter, E.** Cold Temperatures and Increasing Urbanization Favour Melanic Eastern Gray Squirrels on a North American Scale. [Awaiting final coauthor feedback].

Non-Refereed Publications

- 3 **Porter, E.**, Lefevre, K., and Grieves, L. (2025). *Seeking contributions to document the history of the Society of Canadian Ornithologists*. *Picoides*. 38(3): 23.
- 2 **Porter, E.** (2025, August 27). *Bat Roostmates Share More Than Just Space: Exploring the Invisible World of the Gut Microbiome*. College of Biological Sciences SCRIBE Research Highlights, University of Guelph. <https://www.uoguelph.ca/cbs/news/2025/08/bat-roostmates-share-more-just-space-exploring-invisible-world-gut-microbiome>
- 1 **Porter, E.** (2025, August 25). *Sparrow Supermoms: Why Some Female Birds 'Double-Brood'*. College of Biological Sciences SCRIBE Research Highlights, University of Guelph. <https://www.uoguelph.ca/cbs/news/2025/08/sparrow-supermoms-why-some-female-birds-%E2%80%98double-brood%E2%80%99>

Software

- 2 Squirrel Life [Android and iOS mobile application software]. (2021). **Contributors: Porter, E.**, McKechnie, I., Lit, B., Saha, J., Luo, M. X., Norman, Z., Lewis, N., Jacobs, S., Cottenie, K., and Gillis, D. Citizen science data collection tool and user interface to record behavioural, geographic, and temporal data related to squirrels.
- 1 Squirrel Life [Data portal website]. (2021). **Contributors: Porter, E.**, McKechnie, I., Lit, B., Saha, J., Luo, M. X., Norman, Z., Lewis, N., Jacobs, S., Cottenie, K., and Gillis, D. Citizen science data access portal to explore behavioural, geographic, and temporal data about squirrels collected with the Squirrel Life mobile app.

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PRESENTATIONS

Nine invited presentations; Fifteen contributed.

Invited Presentations

Keynote Address

2022 *Curricular Nature-Based Learning in Higher Education to Support Mental and Environmental Health*, 10th Annual Fear Memorial Conference (2022, June 9 – 10), Dalhousie Medical School, virtual; Accredited Professional Development program for excellence in medical education.

Seminars and Symposia

2023 *Meaningful connections through citizen science, creativity, and computer science*. Guelph Institute for Environmental Research Spring Symposium, University of Guelph, Guelph ON.

2022 *Squirrel Life: An Interdisciplinary Venture*. Guelph Institute for Environmental Research Perennial Symposium: Edge Effects, University of Guelph, virtual.

Community Outreach (non-academic science communication)

2023 *Graduate Research Paths in Ecology*, Wildlife Club, University of Guelph, Guelph ON

2022 *Squirrels in the City*, Cambridge Public Library community learning seminar series, Cambridge ON,

2022 *Squirrel Life: Squirrels, science, and students*, Wildlife Club, University of Guelph, virtual

Panel Discussions

2022 Human-environment relationships Webinar, Guelph Institute for Environmental Research, University of Guelph, virtual

2022 Women in STEM Career Panel, University of the Fraser Valley, British Columbia, virtual

2021 Co-organizer and Expert panelist on community science research; Community Science in Canada Symposium, Canadian Society for Ecology and Evolution, virtual

Contributed Presentations

- 15 **Porter, E.**, Jacobs, S., Smith, A., Cottenie, K. (2025, August 13 – 16). Transparent attribution of Canadian citizen science bird data in the peer-reviewed literature. Society of Canadian Ornithologists 2025 Annual Meeting, University of Saskatchewan. Saskatoon SK, CA.
- 14 **Porter, E.**, Jacobs, S., Smith, A., Cottenie, K. (2025, March 21 – 23). Trends in 34 years of Canadian Bird Citizen Science Literature. Ontario Chapter of The Wildlife Society Conference, University of Guelph. Guelph ON, CA. **(Best presentation award)**
- 13 Wise, C., Cottenie, K., **Porter, E.** (2024, June 24 – 27). Winter Bird Count in The University of Guelph Arboretum [oral presentation]. American Public Gardens Association Annual Conference 2024, Boston, MA, USA.
- 12 **Porter, E.**, Dowhaniuk, D., Lipton, M., and Jacobs, S. (2024, April 30). (Re)imagining a Manufactured Ecosystem. [oral presentation]. Graduate Student Symposium, University of Guelph, Guelph ON, CA.
- 11 Wise, C., Cottenie, K., **Porter, E.** (2024, April 5). Sources of Variation in the University of Guelph Arboretum Winter Bird Count: 1980-2024 [oral presentation]. The Arboretum Annual Research Studio, University of Guelph. Guelph ON, Canada.

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- 10 **Porter, E.**, Lewis, N., Robertson, L., Cottenie, K., Jacobs, S. (2023, May 17 – 18). Citizen Science as an Accessible Framework for Experiential Learning [oral presentation]. University of Guelph Teaching and Learning Innovations Conference 2023. Guelph, ON, Canada.
- 9 Lewis, N., **Porter, E.**, Jacobs, S., and Cottenie, K. (2023, March 24 – 26). Spatial Trends in Eastern Grey Squirrels (EGS) [poster]. Ontario Chapter of The Wildlife Society Methods in Wildlife Research 2023 Annual Conference, Peterborough, ON, Canada. **(Best poster presentation award)**
- 8 Wise, C., **Porter, E.**, Cottenie, K. (2023, March 24 – 26). Sources of Variation in the University of Guelph Arboretum Winter Bird Count Data [poster]. Ontario Chapter of The Wildlife Society Methods in Wildlife Research 2023 Annual Conference, Peterborough, ON, CA. **(Best poster presentation award)**
- 7 **Porter, E.**, Luo, M. X., Lit, B., McKechnie, I., Saha, J., Ratra, P., Lewis, N., Norman, Z., Cottenie, K., Jacobs, S., Gillis, D. (2022, March 7 – 9). Squirrel Life [oral presentation]. INTED 2022 16th Annual International Technology, Education and Development, virtual.
- 6 Lewis, N., Porter, E., Byrne, N., McCarthy, A., Raine, K., Earley, C., Jacobs, S. (2022, May 25 – 27). The Urban Squirrel: Assessing the impact of urbanization on behaviour [oral presentation]. Ontario Ecology, Ethology, and Evolution Colloquium 2022 Conference, virtual.
- 5 Lewis, N., Porter, E., Byrne, N., McCarthy, A., Raine, K., Earley, C., Jacobs, S. (2022, March 26). The Urban Squirrel: Assessing the impact of urbanization on behaviour [oral presentation]. The Ontario Chapter of the Wildlife Society 2022 Annual Conference, virtual. **(Best presentation runner-up)**
- 4 **Porter, E.**, Cottenie, K., and Jacobs, S. (2021, May 13 – 14). Scientific and educational benefits of using citizen science as experiential learning for undergraduate students [oral presentation]. Ontario Ecology, Ethology, and Evolution Colloquium 2021 Conference, virtual. **(Best presentation award)**
- 3 **Porter, E.**, Cottenie, K., and Jacobs, S. (2021). Using citizen science for undergraduate student experiential learning [oral presentation]. Canadian Society for Ecology and Evolution 2021 Conference, virtual.
- 2 **Porter, E.**, and Newton, G. (2018). Aerobic exercise as a treatment for major depressive disorder [oral presentation]. Undergraduate Student Research Seminar, University of Guelph, Guelph ON, CA.
- 1 **Porter, E.**, and Bédécarrats, G. (2018). Does rearing environment influence the effect of single nucleotide polymorphisms on the genetics of skeletal traits in laying hens? [poster]. Undergraduate Student Research Seminar, University of Guelph, Guelph ON, CA.

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STUDENT AND RESEARCHER SUPERVISION

Number of Students: 19 (18 Undergraduates, 1 Masters)

- University of Guelph Students: 14
- External to the University of Guelph: 5
 - Shri Guru Gobind Singhji Institute of Engineering and Technology (Nanded, India)
 - Pandit Deendayal Energy University (Gandhinagar, India)
 - University of Electronic Science and Technology of China (Chengdu, China)
 - Vellore Institute of Technology (Vellore, India)

Disciplinary areas of supervision: Wildlife Biology & Conservation, Classical Studies, Art History, Culture & Technology, Bio-medical Sciences, Software Engineering, Business, Environmental Science, Software Engineering, Computer Science, Computer Engineering, Data Science

TEACHING ACTIVITIES

Guest Lectures – Invited (University of Guelph)

Nov. 2024: *Pathways to Graduate School in Ecology*. Integrative Vertebrate Biology (ZOO*4910) with Dr. Pauline Kosmal, University of Guelph.

Oct. 2024: *Citizen Science as a Way of Knowing*. Advanced Philosophy of the Environment (PHIL*4040) with Dr. Stefan Linquist, University of Guelph.

Mar. 2022: *Squirrels as a Gateway to Nature Connectedness*. Natural History of the Great Lakes (ENVS*1040) with Dr. Jonathan Schmidt and Dr. Andrew Young, University of Guelph.

Course Delivery and Training

Jan. 2021 – Apr. 2021: BIOL*1070 – Discovering Biodiversity, University of Guelph, Guelph ON. **Role: Course Co-Coordinator.** Part of the Course Coordination team formed ad hoc when the original course coordinator had to unexpectedly step back.

Mar. 2019 – Apr. 2019: Junior Naturalists Outdoor Education Program, University of Guelph Arboretum, Guelph ON. **Role: Outdoor Educator.** After school outdoor education program for children ages 6-12. Developed and led nature-based education programming for the University of Guelph Arboretum.

Jan. 2018 – Jan. 2024: Raptor Husbandry and Behaviour, Wild Ontario, Guelph ON. **Role: Team Lead and Trainer.** Primary trainer for 7 trainees. Responsible for coordinating training schedules, conducting evaluations, communicating progress to the Program Director, and ensuring preparedness for the final evaluation to confirm successful completion and transfer from trainee to Raptor Handler.

Jul. 2013 – Dec. 2016: Lifesaving Society Aquatic Leadership Courses, City of Cambridge, Cambridge, ON. **Role: Instructor.** Trained over 50 participants in essential aquatic lifesaving skills, including emergency water rescues and first aid.

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Dec. 2011 – Dec. 2016: Canadian Red Cross Aquatics Courses, City of Cambridge, Cambridge, ON.

Role: Instructor. Instructed over 3,000 students. Often assigned to teaching adult learners and private classes because of demonstrated success, innovative teaching strategies, and excellent communication with patrons.

Course Review and Development (*University of Guelph*)

Jan. 2025 – Apr. 2025: **Course Revision**, BIOL*1500 – Humans in the Natural World

Revision of content and online learning modules to improve relevance and timeliness of course content.

Jan. 2023: **Course Revision**, BIOL*1070 – Discovering Biodiversity

Updated experiential learning project (Squirrel Life Project) modules based on student feedback.

Aug. 2022: **Course Review**, BIOL*3660 – Internship in Biological Science

Critically reviewed assessment instructions and evaluation criteria and provided recommendations for evidence-based improvement.

Dec. 2021 – Jan. 2022: **Course Development**, BIOL*1070 – Discovering Biodiversity

Developed novel, evidence-based experiential learning modules and assessments to deliver core curriculum.

Sep. 2020 – Dec. 2020: **Course Revision**, BIOL*1070 – Discovering Biodiversity

Adapted course modules and revised content for virtual delivery during the COVID-19 pandemic.

Teaching Assistantships (*University of Guelph*)

Introduction to One Health, ONEH*1000 (*Jan. – April 2026*)

Humans in the Natural World, BIOL*1500 (*Sep. – Dec. 2022; Sep. – Dec. 2023; Jan. – Apr. 2025*)

Discovering Biodiversity, BIOL*1070 (*Sep. – Dec. 2020; Jan. – Apr. 2021; Sep. – Dec. 2021; Jan. – Apr. 2023*)

Internship in Biological Science, BIOL*3660 (*May – Aug. 2022*)

Ecology, BIOL* 2060 (*May – Aug. 2022*)

Tutoring

Sep. 2015 – Apr. 2016. Private instruction to undergraduate students to improve academic performance in biology, math, physics, chemistry, biochemistry, and English language skills. Worked with students to identify challenges, set goals, and implement solutions.

Attendance at Teaching Conferences and Workshops

- **Graduate Student University Teaching Program**, 2021 (Aug 23 – 26), University of Guelph – Office of Teaching and Learning
- **University Teaching Foundations**, 2020-22, University of Guelph – Office of Teaching and Learning
- **Scholarship of Teaching and Learning Hackathon Day**, Oct. 2020, virtual, Society for Teaching and Learning in Higher Education. (International). Collaborated with an interdisciplinary working group to develop novel research solutions for the Scholarship of Teaching and Learning in a virtual/hybrid learning context (during the COVID-19 pandemic).

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CREATIVE SCHOLARSHIP

Ary Gallery Exhibition Catalogues

Dowhaniuk, D., & Porter, E. (Eds.). (2025). *Manufactured Ecosystems Program Guide and Education Booklet*.

<https://www.manufacturecosystems.com/manufactured-ecosystems-exhibition>

Art Gallery Exhibitions Curated

Jun. 2025 – Aug. 2025

Manufactured Ecosystems, **Role: Co-curator** with Dave Dowhaniuk, Alyssa Ponte, Annika Yu, Hanah Hiemstra, and Sara Jandrew. Zavitz Gallery, University of Guelph, Guelph, ON.

Immersive exhibition exploring speculative futures at the intersection of art, technology, and nature to address the climate crisis. Served as the public-facing centrepiece of an International research initiative supported by the Government of Canada's New Frontiers in Research Fund. Featured interactive installations, speculative prototypes, and imaginative storytelling envisioning how humanity might adapt to ecological collapse through nature-inspired technologies.

VOLUNTEERING

Alan Wormington Fall Bird Count [Cambridge, Team Leader for Area 7]

Hamilton Naturalists' Club, Nov 2025

Ontario SwiftWatch Surveyor

Birds Canada, May – Sep 2025

- Monitor local Chimney Swift habitat to collect data on habitat use, and presence and abundance of Chimney Swifts
- In 2025 I conducted 25 Chimney Swift surveys spanning 11 Cambridge sites for SwiftWatch, updating data for the first time in 14 years for some sites.

Monarch Butterfly Tagger

Monarch Watch, Aug – Sep 2025

Great Lakes Marsh Monitoring Program – Bird and Amphibian Surveyor

Birds Canada, Mar-Jul 2025

- Established a new survey route [Barrie's Lake] and surveyed ON091
- Monitored local wetlands to determine the presence and abundance of marsh-dependent avian and amphibian species using visual, auditory, and callback surveys.

Odonate Inventory Volunteer

University of Guelph Arboretum, Jul 2025

- Surveyed the Arboretum for dragonflies and damselflies, captured, identified and released

Breeding Bird Survey Volunteer [Emily Park, Route #68417]

Environment and Climate Change Canada, May 2025

- Conduct standardized early-morning bird point count surveys along a designated roadside transect during the peak breeding season.
- Identify bird species by sight and sound, record observations at pre-determined stops, and adhere to established protocols to ensure data consistency and scientific integrity.

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Cambridge Christmas Bird Count [Team Leader for Area 3]
Birds Canada, Dec. 2024-2025

Guelph Christmas Bird Count [Area 8]
Birds Canada, Dec. 2023

Arboretum Winter Bird Count
University of Guelph Arboretum, Jan 2023-present

Project FeederWatch Volunteer
Birds Canada, 2023-present

- Participate annually (2 days per week from November to April) in Project FeederWatch by following a citizen science bird data collection protocol and recording observations with a citizen science mobile application.

Global PPE Litter Tracking Surveyor
Jul. to Oct. 2021

- Worked with National Geographic Explorers (Justine Ammendoila and Jackie Saturno) to conduct personal protective equipment (PPE) surveys in public outdoor spaces in Guelph, ON to contribute to the international initiative: "Global plastic pollution during the pandemic: Tracking personal protective equipment (PPE) in the time of COVID-19".

SERVICE ROLES

Executive Committees

Jan. 2026 – present: **Society of Canadian Ornithologists – Student Affairs** (National).
 Co-chair and founding member.

Sep. 2025 – present: **Society of Canadian Ornithologists – Website Committee** (National).

Sep. 2017 – Apr. 2018: **Poultry Club**, Ontario Agricultural College, University of Guelph. (Institutional).
 Elected to the Executive Council to serve as the Breeding Program Coordinator.

Sep. 2016 – Apr. 2017: **Circle K International – Guelph Chapter Executive Council**, Kiwanis Canada, University of Guelph. (Institutional).
 Co-founding member. Elected to the Executive Council of the Chapter to serve as VP Fundraising.

Committee Member

Sep. 2024 – present: **Admissions and Progress Committee**, Board of Graduate Studies, University of Guelph, Guelph, ON. (University Service).

Responsible for reviewing graduate admissions and student progress policies, adjudicating special requests and cases of unsatisfactory process. Voting member. University Senate subcommittee.

Sep. 2025 – Nov. 2025: **Invited Speaker Organizing Committee**, Department of Integrative Biology, University of Guelph, Guelph ON. (Department committee).

Sep. 2021 – Aug. 2022: **Teaching and Experiential Learning Committee**, Department of Integrative Biology, University of Guelph, Guelph ON. Department committee.

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Community Outreach Activities

2016-2023 Wildlife and Conservation Educator, Wild Ontario, Guelph ON. Delivered educational programming on avian ecology and conservation science for diverse audiences (K to University, community events), working with non-releasable birds of prey.
2018-2023: Team Lead. Mentored volunteers in developing science communication skills.

2015 Wildlife Rehabilitator, Wildlife Haven Waterloo, Waterloo ON
 2016 Vida Veterinary Program Community Education and Outreach, Costa Rica
 2015 Wildlife and Conservation Educator, ScienSational Snakes, Guelph ON

Communications

2023-2025 Manufactured Ecosystems, education content & research creation.
 2022 Gardening Educational Content Creator, Guelph ON
 2020-2021 Social media educational content creator, Wild Ontario

Event Administration

Conference – Facilitation

Aug. 2021: Conference: Session Facilitator. **Graduate Student University Teaching Days**, Office of Teaching and Learning, University of Guelph, Guelph, ON. (Institutional).

Facilitated sessions during a Teaching Assistant networking event, supporting connection and knowledge exchange between new and experienced TAs. Led introductions, guided discussion, and fostered peer mentoring across diverse teaching formats and colleges.

May 2021: Conference: Session Moderator / Facilitator. **College of Biological Science Graduate Student Symposium**. (Institutional) (College Service). University of Guelph, Guelph, ON.

Facilitated 2 talk sessions, Responsible for keeping the session on track, guiding discussions, and providing technical support to presenters.

Conference - Organizer

Jan. 2021 – May 2021: Conference: Organizing Committee. **College of Biological Science Graduate Student Symposium**. (Institutional) (College Service). University of Guelph, Guelph, ON.

Co-organizer. Secured funding and sponsors; managed the call for abstracts, and recruited judges.

Contributed to session organizing and coordinated plenary speakers. Provided logistical support for presenters and attendees to ensure a seamless and impactful experience.

Conference – Session Organizer

May 2021: Conference: Symposium Organizer. **Community Science in Canada**, Canadian Society for Ecology and Evolution Conference.

Co-organizer with Carly Ziter, Pierre Chuard, Erin Smith, and Victoria MacPhail. Coordinated session theme, created resources, invited participants, and facilitated working group discussion.

Conference – Communications Coordinator

May 2021: Conference: Communications Lead. (Institutional Service), Graduate Student Symposium, **College of Biological Science**, University of Guelph. (College Service).

Led a three-day social media campaign on the College of Biological Science's Twitter (1,178 followers) and Instagram (886 followers) to promote conference programming and graduate student research, increasing audience engagement through live event coverage.

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Seminar Series – Organizer

Jan. 2021 – Feb. 2021: Organizer, **Exploring Wildlife Seminar Series** (National).

Organized and hosted a virtual seminar series about animal signs, adaptations, and tracking for the community. Over 80 attendees from across Canada.

Jun. 2025 – Jul. 2025: Organizer, **Artist Talk Seminar Series**. (International).

Organized and hosted a virtual artist talk series for community members and researchers. Over 50 attendees from around the world.

Nov. 2023: Organizer, **Shaping Equitable Spaces Seminar**. (International)

Organized and hosted a virtual seminar and roundtable discussion about equitable practices in knowledge creation and research spaces for a group of 50 international research collaborators.

Oct. 2023 – Apr. 2024: Organizer, **Interdisciplinary Knowledge Sharing Seminar Series**. (International).

Organized and hosted a virtual seminar series with speakers from engineering, ecology, education, art history, literature, and more for a group of 50 international research collaborators.

Workshops – Organizer

May. 2024 – Aug. 2024: Organizer, **Summer Interdisciplinary Research Workshop Series**. (Institutional).

Organized a summer workshop series for a group of 15 early-career researchers and faculty members, coordinating facilitators, scheduling, and dissemination of workshop resources. Topics included environmental sciences, population medicine, art history, software engineering, and more.

Jun. 2025: Organizer, **Art Creation Workshop**. (Local). University of Guelph, Guelph, ON.

Organized and hosted a workshop led by Canadian visual artist Yulia Shtern, creator of the internationally exhibited *Magical Zoo* sculpture series, engaging 15 participants in a hands-on art experience.

Aug. 2024 – Feb. 2025: Organizing Committee, **Art-Science Collaborative Workshop Series**. (International).

Co-organized and hosted 2 workshops and asynchronous collaborative processes that brought 30 scientists, artists, writers, and other collaborators together to imagine the future of technology in the context of a climate collapse. Part of a research project funded by the Government of Canada's New Frontiers in Research Fund.

Community Events – Organizer

Mar. 2022: Organizer, **Public Event – Squirrel Life Project Community Launch**, University of Guelph, Guelph, ON. (Local) (Community Service).

Sole organizer Over 150 people attended.

Community Events – Facilitator

Aug. 2019: Event Administration (Community), **Public Event – Guelph Bug Day**, University of Guelph, Guelph, ON. (Local) (Community Service).

2017 – 2023: **Public Event – College Royal**, University of Guelph.

Annually during College Royal in March. Plan and deliver educational presentations on bird behaviour and ecology, entomology, and poultry husbandry for the broad public as a part of various volunteer groups at the University of Guelph. Interact with members of the public to answer questions.

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Assessment and Review Activities

Conference Review Activities

- 2023: **Reviewer: Conference Proposals**, Teaching and Learning Innovations Conference. 3 proposals. (Regional).
- 2021: **Reviewer: Conference Proposals**, Ontario Ecology, Ethology, and Evolution Colloquium. 7 proposals. (Regional).

Undergraduate Assessment

Nov. 2024: **Evaluator**, Computer Science Project Demonstrations. CIS*3750. 4 teams. (Institutional).
 Oct. 2024: **Evaluator**, Computer Science Project Demonstrations. CIS*3750. 4 teams. (Institutional).
 Dec. 2021: **Evaluator**, Internship Student Presentations, virtual. BIOL*3660. 6 presentations. (Institutional).
 Nov. 2021: **Evaluator**, Computer Science Project Demonstrations, virtual. CIS*3750. 4 teams. (Institutional).
 Oct. 2021: **Evaluator**, Computer Science Project Demonstrations, virtual. CIS*3750. 4 teams. (Institutional).
 Nov. 2020: **Evaluator**, Research Project Poster Presentations, virtual. IBIO*4521. 3 posters. (Institutional).

Workshop Evaluation

Aug. 2025: **External evaluator**, The Arboretum – University of Guelph. (Professional Service).
 Evaluation of the first offering of the *Galls of Southern Ontario* workshop delivery, content, and educational resources.

Jul. 2025: **External evaluator**, The Arboretum – University of Guelph. (Professional Service).
 Evaluation of the *Dragonflies and Damselflies* workshop delivery, content, and educational resources.

Jul. 2025: **External evaluator**, The Arboretum – University of Guelph. (Professional Service).
 Evaluation of the *Mysterious World of Moths* workshop delivery, content, and educational resources.

Jul. 2025: **External evaluator**, The Arboretum – University of Guelph. (Professional Service).
 Evaluation of the *Identification and Biology of Butterflies* workshop delivery, content, and educational resources.

Other Service Activities

Sep. 2016: Campus Ambassador/Orientation Volunteer, University of Guelph, Canada. (Local) (University Service). Welcome and support new undergraduate students during Orientation Week by assisting with event logistics, building connections within academic communities, and providing guidance to students on campus.

2024-present SCRIBE Writer, College of Biological Science, University of Guelph

2020-present College of Biological Sciences Undergraduate Mentorship Program, University of Guelph

Sep. 2016 – Dec. 2016: Peer Helper, **Wellness Education Centre**, University of Guelph. (Institutional service). Collaborated on a team to promote student health and wellness through workshops, events, and outreach on topics such as mental health, sexual health, and substance use. Supported education by developing resources, creating communication materials, and assisting students in accessing campus wellness supports.

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MEDIA

June 2025 [Manufactured Ecosystems: Art Merges with Science to Rethink Climate Adaptation](#)

Broadcast Interviews:

- 11 interviews covering the launch of the Squirrel Life Project and associated community events. Included Local, Regional, and National radio and TV interviews (live and recorded); guest on a podcast to talk about human-nature connections and squirrel biology.

RESEARCH GRANTS AND AWARDS

Total funding ~\$189,645 CAD

Research Grants

Title and Role	Funding Organization	Amount	Funding Period
Title: The Squirrel Life Project: Community Expansion Role: Principal Applicant	Chantiers jeunesse – Social Impact Accelerator Program	\$5,000 CAD	May – Dec. 2023
Title: Canada Graduate Scholarship – Doctoral (CGS D) Role: Principal Applicant	Natural Sciences and Engineering research Council of Canada (NSERC)	\$113,333 CAD	Sep. 2022 – Dec. 2026
Title: Backyard Nature Role: Principal Applicant	Chantiers jeunesse – Social Entrepreneurship Program	\$2,500 CAD	May – Aug. 2022
Title: The Squirrel Life Project: Community Initiative Role: Principal Applicant	Chantiers jeunesse – Social Entrepreneurship Program	\$2,500 CAD	Nov. 2021 – Apr. 2022
Title: The Educational and Scientific Benefits of Using Citizen Science Projects as Experiential Learning Opportunities for Undergraduate Students Role: Co-applicant	Associate Vice-President Academic Scholarship of Teaching and Learning Grant, University of Guelph	\$10,000 CAD	May 2021 – Apr. 2022

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Title: Queen Elizabeth II Graduate Scholarship in Science and Technology Role: Principal	Ministry of Training, Colleges and Universities; University of Guelph Funding Type: Scholarship	\$15,000 CAD	May 2021 – Apr. 2022
Title: Squirrel Life: Citizen and Computer Science Role: Co-applicant	Guelph Institute for Environmental Research (GIER) Funding Type: Grant	\$15,000 CAD	Jan. 2021 – Dec. 2021
Title: The Squirrel Life Project: Community Outreach Role: Principal Applicant	Student Life Enhancement Fund, University of Guelph Funding Type: Grant	\$2,200 CAD	Jan. 2021 – Apr. 2021
Title: Wild Ontario Experiential Engagement Event Role: Co-applicant	Student Life Enhancement Fund, University of Guelph Funding Type: Grant	\$1,000 CAD	Aug. 2019
Title: Entrance Scholarship Role:	University of Guelph Funding Type: Scholarship	\$3,000 CAD	Sep. 2014
Title: Western Scholarship of Excellent Role:	Western University Funding Type: Scholarship	\$10,000 CAD	Sep. 2014 <i>Declined</i>
Title: Harold W. Lee Memorial Scholarship Role:	Preston High School Funding Type: Scholarship	\$1,000 CAD	Sep. 2014
Declined Research Grants			
Title: Northern Scientific Training Program Role: Principal Applicant	Polar Knowledge Canada (POLAR) Funding Type: Research Grant	\$3,270 CAD	March 2025 <i>Declined due to travel scheduling conflicts.</i>
Title: Ontario Graduate Scholarship (OGS) Role: Principal Applicant	Ministry of Training, Colleges and Universities; University of Guelph Funding Type: Scholarship	\$15,000 CAD	March 2022 <i>Declined due to NSERC</i>

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Awards

Award	Organization	Amount	Award Date
Braithwaite Conference Travel Grant	University of Guelph	\$2,818 CAD	July 2025
Best Oral Presentation	Ontario Chapter of the Wildlife Society Annual Conference	\$250 CAD	March 2025
Braithwaite Conference Travel Grant	University of Guelph	\$392 CAD	March 2025
Best Submission	Health Science Inquiry Journal	\$500 CAD	May 2022
First Place, 3-minute Thesis Competition	Ontario Ecology, Ethology, and Evolution Colloquium Annual Conference	\$75 CAD	May 2021
Academic and Community Involvement Bursary	Kin Canada	\$1,000 CAD	June 2015
Governor General's Bronze Medal for Academic Excellence	Preston High School	NA	2014
The Canada Cord	Girl Guides of Canada	NA	2010
Lady Baden Powell Award	Girl Guides of Canada	NA	2008

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

Identification of Dragonflies and Damselflies (7 hours)

July 2025, University of Guelph Arboretum, Trainer: Chris Early, Arboretum Interpretive Biology

Moth Identification (6 hours)

July 2025, University of Guelph Arboretum, Trainer: Andalyne Tofflemire, Cambridge Butterfly Conservatory

Identification and Biology of Butterflies (7 hours)

July 2025, University of Guelph Arboretum, Trainer: Andalyne Tofflemire, Cambridge Butterfly Conservatory

Bird Sounds Series (8 hours)

March – April 2025, virtual, Trainer: Chris Earley, University of Guelph Arboretum Interpretive Biologist

Mar. 2025: Introduction to VHF Telemetry

Guelph ON, Trainer: Matt Purvis, Fleming College

Jun. 2025: The Fundamentals of OCAP® (Ownership, Control, Access, Possession of First Nations Data)

First Nations Information Governance Centre

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Exploring One Health

2023, virtual, University of Guelph – One Health Institute

Tri-Council Course on Ethical Conduct for Research (TCPS 2: CORE-2022)

2021, virtual, Tri-Council Panel on Research Ethics

Poultry Industry Council Artificial Insemination Certificate Level I

2017, Guelph ON, Trainer: Innes Wilson, Arkell Poultry & Equine Research Station

Poultry Industry Council Artificial Insemination Certificate Level II

2018, Guelph ON, Trainer: Innes Wilson, Arkell Poultry & Equine Research Station

Social Entrepreneurship Training Program (20 weeks)

2021-22, Mentor: Mathieu Constantin, Chantiers Jeunesse Outreach and Development Officer

Indigenous Canada (12 weeks)

2021, virtual, University of Alberta

Science Communication Toolkit: Impactful Presentations (2 hours)

2021, virtual, Trainers: Keriann McGoogan and Virginia Capmourteres

Universal Instructional Design Training: Making Education Accessible

2021, virtual, University of Guelph

Standard First Aid, Emergency First Aid, CPR/AED Level C

2015-2023

Graduate Research and Project Management Course (11 hours)

Jan – Mar 2022, virtual, University of Guelph – Office of Research

R programming Course: Special Topics in Ecology and Behaviour (12 weeks)

2021, virtual, Instructor: Dr. Karl Cottenie, University of Guelph

Skills for Science Communication: Graphic Design for Infographics (1.5 hours)

2021, virtual, Trainer: Carlea Blight, University of Guelph Library, Media Studio Technician

Grad Pathways: Professional Skills Development Course

2020-22, virtual, University of Guelph Office of Graduate & Postdoctoral Studies

Skills for Research Impact (9 hours)

2020, Guelph ON, Training Organizations: Community Engaged Scholarship Institute, Ontario Agri-Food Innovation Alliance, and Research Innovation Office

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Data Skills Immersion Program (4 days), 2020, Guelph ON.

Principles of Belonging: Anti-Oppression and Anti-Racism (1 hour)
2020, virtual, Office of Diversity and Human Rights (University of Guelph)

More Feet on the Ground, 2017, Guelph ON, Centre for Innovation in Campus Mental Health

safeTALK suicide prevention training, 2017, Guelph ON, LivingWorks

Accessible Service Provision University of Guelph, 2016, virtual, University of Guelph

Wild Ontario Experiential Learning Program:

Raptor Husbandry and Training; Environmental Education (20 weeks)

2016, Guelph ON, Trainers: Jenn Bock and Kyle Horner, Wild Ontario (University of Guelph)

Jan. 2016: **Animal User Training Program**

University of Guelph, Guelph ON.

Feb. 2014: **Supervisor Occupational Health and Safety Awareness Certification**

Ontario Ministry of Labour, Training and Skills Development. Cambridge, ON.

Jan. 2014 – Jan. 2016: **Pool Operator – Level 1 Certification**

Lifesaving Society. Cambridge, ON.

Training in swimming pool operations and safety management, covering facility operation, maintenance practices, and adherence to provincial health and safety regulations.

Jun. 2013: **Aquatic Supervisor Certification**

Lifesaving Society. Cambridge, ON.

Training to assume deck-level supervisor responsibilities for overseeing instructional and recreational programs and day-to-day operations at aquatic facilities.

Jun. 2013 – Jan 2017: **Swim Instructor Certification**

Lifesaving Society. Cambridge, ON.

Jun. 2013 – Jan. 2017: **Lifesaving Instructor Certification**, Lifesaving Society. Cambridge, ON.

Jun. 2013 – Jan. 2017: **Emergency First Aid Instructor Certification**, Lifesaving Society. Cambridge, ON.

Aug. 2012 – Dec. 2017: **National Lifeguard Certification**, Lifesaving Society. Cambridge, ON.

Dec. 2011 – Dec 2017: **Water Safety Instructor Certification**, Canadian Red Cross. Cambridge, ON.

Oct. 2011: **Principles of Health Child Development**, HIGH FIVE ®. Cambridge, ON.

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PROFESSIONAL AND OTHER MEMBERSHIPS

Feb. 2026 – present, Ontario Field Botanists. (Regional).

Jan. 2026 – present, Ducks Unlimited Canada Campus Club – University of Guelph. (Institutional).

Feb. 2025 – present, Society of Canadian Ornithologists. (National).

Feb. 2025 – present, Ontario Bird Banding Association. (Regional).

Jan. 2024 – present, Galt Horticultural Society. (Local).

Oct. 2023 – present, Ontario Field Ornithologists. (Regional).

Mar. 2022 – present, Ontario Chapter of the Wildlife Society. (Regional).

Jan. 2022 – Dec. 2024, Guelph Horticultural Society. (Local).

Jun. 2021 – Dec. 2022, Canadian Society for Ecology and Evolution. (National).

Nov. 2020 – Jun. 2023, Guelph Institute for Environmental Research – Early Career Affiliate. (Institutional).

Sep. 2016 – Apr. 2018, Poultry Club, University of Guelph. (Institutional).

Sep. 2016 – Apr. 2017, Circle K International – Guelph Chapter. (Institutional).

ADDITIONAL SKILLS

- Data analysis (R/RStudio)
- GitHub
- Website tools: Squarespace, WordPress, Quarto
- Project Management skills
- NVivo Qualitative Analysis Software
- Qualitative research methods (interviews, survey design, thematic analysis)
- Literature meta-analysis